duction band may also be defined as the lowest unfilled energy band at 0K temperature.

3.6.3 Classification as Insulator, Semiconductor and Conductor

On the basis of forbidden energy bands, all the solids can be classified into insulator, semiconductor and conductor, as given below :

(i) **Insulators :** In the case of insulator, the forbidden energy band is very wide [Fig. 3.6 (i)] and so the electrons cannot jump from valence band to conduction band. Moreover in insulators, the electrons are very tightly bound to the parent atom, e.g., in case of an insulator like glass, the valence band is completely full at 0K and the energy band gap is ~ 10 eV. Even if there is very high electric field, the electrons cannot jump from valence band to conduction band. However, if very high energy is supplied, an electron may be able to cross the forbidden band. If the temperature is increased, then some electrons may jump the forbidden band and this is why certain materials which are insulators become, good conductors at high temperatures. The normal range of resistivity of insulators is ~ 10^7 ohm.

(ii) **Semiconductors :** These are the solids whose electrical properties lies in between those of conductors and insulators. In this case the forbidden band is very small. [Fig. 3.6 (ii)]. Examples

of semiconductors are germanium, silicon etc. On germanium, forbidden bands is 0.7 eV and in silicon it is 1.1 eV. Conduction band has no electrons and valence band is completely filled at 0K. If some energy is supplied then electron jumps from valence band to conduction band. In the case of semiconductors, conductivity is ~ 10^2 mho – m. As temperature is increased, more valence electrons cross over to the conduction band and conductivity increases.

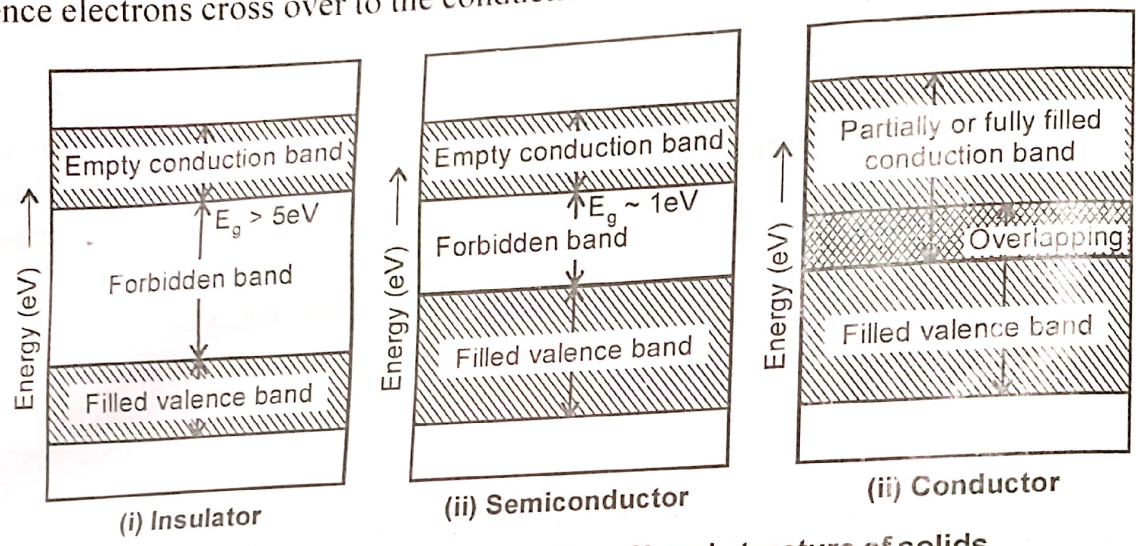

Fig. 3.6 : Schematic representation of band structure of solids

(iii) Conductors : Conductors are characterised by high electrical conductivity. These are the solids with no forbidden energy band [Fig. 3.6 (iii)] and the valence and conduction bands overlap each other. Hence, large number of free electrons are available for electric conduction and electrons from valence band freely enter into the conduction band. Now since in the case of conductors, there is no forbidden band, so holes are not created at all and the total flow of current is due to the motion of electrons only. Example Silver, Copper, Iron etc.

3.7 FERMI DISTRIBUTION FUNCTION *f* (E)

The fermi distribution function, also called the fermi function gives the probability of occupancy of energy levels by fermions in terms of fermi energy, temperature and Boltzmann constant.

Fermions are half integer spin particles like electrons which obey Pauli exclusion principle.

Consider a system of electrons in thermal equilibrium at a temperature T K. The probability $f(E)$ of an electron occupying a particular energy level (state) E is given by

$$f(E) = \frac{1}{1+e^{(E-E_F)/k_BT}} \qquad ...(3.8)$$

where $f(E)$ is the fermi function.

E_F is the fermi energy (also called fermi level)

k_B is the Boltzmann constant

T is the absolute temperature.

Thus, a system is characterised by its temperature and its fermi energy E_F.

level and so on, till all the electrons of the metal are accommodated as shown in figure.

The uppermost filled energy level is referred as fermi level and the energy of the electrons in the fermi level at absolute zero degree is called the fermi energy (E_F).

3.9 SEMICONDUCTORS

Semiconductors are solid materials having electrical resistivity in between those of conductors and insulators. In electrical conductivity of semiconductors, both electrons and holes are the charge carriers and their electrical conductivity varies with temperature and impurity. This gives an edge to the application of semiconductor. A semiconductor can be a conductor as well as an insulator at different conditions. At 0 K, all semiconductors behave as insulators because their valence bands are filled and conduction band is empty. Semiconductors have negative temperature coefficient of resistance and its resistance decreases with increasing temperature.

Semiconductors are used in manufacturing of electronics devices and integrated circuits. Commonly used semiconductors are Si, Ge, GaAs and InP.

3.9.1 Types of Semiconductors

Semiconductors can be classified into following categories.

(i) **Elemental and compound semiconductors :** Based on the number of constituting elements.

(ii) Direct and indirect band gap semiconductors : Based on the type of energy emission.

(iii) Intrinsic and extrinsic semiconductors : Based on purity.

Elemental semiconductors are composed of single species of atoms and are found in Group-IV of periodic table. **Examples :** Si and Ge.

Compound semiconductors* are composed of two or more different species of atoms. Further, compound semiconductors consisting of two, three and four elements are called binary, ternary and quarternary compounds respectively.

Examples :

Binary compounds :	GaAs, GaP, InP, InAs, InSb (Group-III + V)
	Zns, CdS, CdTe, ZnTe (Group II + VI)
Ternary compounds :	GaAsP, AlGaAs
Quarternary compounds :	AlGaAsP, InGaAsP

Direct band gap semiconductors emits# light energy (photons; $h\nu$) and used to make LED's and lasers of different colours. This process is known as radiative recombination or spontaneous emission. **Examples :** GaAs, AlAs, InP

Indirect band gap semiconductors emits# heat energy (phonon) and dissipated within the material. This process is known as non-radiative recombination.

Intrinsic semiconductors are pure semiconductor materials and possess poor conductivity at room temperature solely due to the breaking of few covalent bands by thermal excitation of electrons from valence to the conduction band leaving behind a vacancy in valence band known as a "hole". The holes may be filled by another electrons in valence band. The elemental and compound semiconductors can be intrinsic semiconductors.

Extrinsic semiconductors is an impure form of an intrinsic semiconductors (Si or Ge) by adding a small quantity of impurity atoms to improve its conductivity. The conductivity depends on the number of impurity atoms known as dopant. Based on element of doping, the extrinsic semiconductors further categorised in two types.

(a) *n*-type semiconductor ; formed by doping atoms (As, P, Sb) having five valence electrons

(b) p-type semiconductors ; formed by doping atoms (Ga, In, B) having three valence electrons.